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Intra-EU Aggregate Trade and The Maastricht Treaty

Abstract:

Using data from a number of EU member states, I look at the impact the Maastricht Treaty had on intra-EU trade using a fixed effects difference in differences framework. I am able to separate trade data by industry, and this allows me to use the coal and steel industries as a control group. Prior to the Maastricht Treaty these industries were already protected by an international trade agreement. Although the effects of contiguity and other trade barriers have been thoroughly studied, the literature on the impact the Maastricht Treaty had on intra-EU trade is surprisingly sparse. I find the treaty led to an increase in aggregate trade among goods not already protected from tariffs among the largest European Union member countries. The results confirm boilerplate models of international trade and suggest Brexit could have significant impacts on trade flows of the United Kingdom and other EU countries.

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Keywords: International Trade

Introduction:

On February 7, 1992 members of the European Communities signed the Maastricht Treaty, forming the European Union. This research explores the Maastricht Treaty's impact on trade across the European Union, focusing on aggregate trade between the United Kingdom, Spain, Italy, Germany, and France. I will compare aggregate trade between these countries before and after the treaty. the EU is an enormous economic entity. It created enormous economic change and reshaped the economic role Europe holds with the rest of the world. On top of this, my data set includes the United Kingdom (UK), and by recognizing the impact the Maastricht Treaty had on trade in the UK and other large European countries we might better understand the consequences of the UK leaving the EU.

The principle goal of the Maastricht Treaty was to further European integration. The treaty accomplished this through a number of ways, one of which was making the member states economically dependent on one another. The Maastricht Treaty created a single market where goods, people, services, and capital could move freely across borders and was the realization of the decades long attempt to create a more integrated and economically efficient Europe. The treaty created a European Union consisting of three pillars: The European Communities, Common Foreign and Security Policy, and Cooperation in Justice and Home Affairs. Even before these pillars were established by the Maastricht Treaty, there had already existed economic communities in Europe. The major such community that had already been established was the European Economic Community (EEC). After the treaty, it was absorbed into the first pillar, the European Communities. The EEC, formed in 1957, had neither the number of signatories, nor the broad reaching power that the Treaty of Maastricht granted the EU. The most significant institution in the EEC for my purpose was the European Coal and Steel Community. This community would also become a part of the Treaty of Maastricht in 1993.

My model shows that the creation of the EU decreased trade barriers and led to an increase in aggregate trade between member states. Using a fixed effects difference in differences model, I control for both entity and time fixed effects and identify changes in aggregate trade that occurred among industries impacted by the creation of the EU. I find that the implementation of the Maastricht Treaty in 1993 (1994 fully) led to a significant increase in tradeable goods that had not been previously protected by international trade agreements. The results also show that goods that had been part of previous trade agreements (coal and steel) also increased in aggregate trade, although not to the same extent as the other goods did. Finally, removing Germany from the equation still gives me the same result as before, indicating that Germany, the country that participates in intra-EU trade the most, was not the lone driver of my results.

After contextualizing my own research in both past scholarship and broader economic theory and my own research's relevance to the economic theory of international trade in a region with reduced economic barriers, I will move to discuss the data I have chosen and why I picked the countries, statistical model, and time period that I did. Lastly, I will provide my results and my conclusion on the effect of the Maastricht Treaty and its impact on trade across the European Union.

Literature Review:

Being unable to find papers discussing the direct impact the Maastricht Treaty had on intra-EU trade, I will instead discuss existing literature relating to the economic effects resulting from the creation and removal of trade barriers. The two areas of study that are most important to this paper are: 1) the impact on international trade when there is a reduction in economic barriers and, 2) the impact of the EU on member states in terms of trade barriers. Both tariff, and non-tariff trade barriers (NTBs) exist throughout international trade and are considered to be quite

large. Tefler (1993) found sizeable non-tariff barriers while Harrigan (1993) found sizable tariff barriers Hayeman (2003) found sizeable tariff barriers and slightly less sizeable but still large NTBs. Not surprisingly, when these barriers are reduced, trade between partners opens up significantly (Hayeman 2003). One example of NTBs are borders. Chen (2004) discusses the border effect on trade. Specifically, she considers this effect among European Union member countries. Even though borders reduce trade in general, she concludes that in the case of EU countries, their market integration leads to a reduction in border effects on trade, and predicts an increase in intra-European trade. Clearly, a reduction in trade barriers leads to an increase in trade, but was the market integration in the EU also a reduction of barriers to trade? Chen's paper seems to imply so.

The creation of the EU had a number of immediate effects, including the recognition of standards and the abolition of border controls. By reducing border effects through these measures, intra-EU trade increased (Chen 2004). The abolition of barriers and integration brought forth by the creation of the EU was so effective that we see backlash against it today, as countries in the EU begin to create more NTBs by using national regulations to make their own producers more competitive in the single market. Perhaps the most extreme instance of this is Britain leaving the EU. Countries employ NTBs to support their own national interest over the collective interest of the EU (Guimaraes, 2010). This slight closing off of the market is only now happening because previous barriers to trade were reduced so dramatically when the EU was created in 1993. Throwing up trade barriers seems counterproductive Europe as a whole, as the single market created and continues to have a "large positive impact on trade" (Fournier, 2015). In summary, we now know that a reduction of trade barriers between countries results in increased trade between partners (Hayeman, 2003) and that the creation of the EU resulted in a decrease in trade barriers and greater economic integration for member countries, causing a

greater increase of trade between partners. This is only now beginning to see push back by member states as they seek to regain more control over their trade. Given both of these observations, I predict that the creation of the EU resulted in a decrease in barriers to trade and an increase of aggregate intra-EU trade.

The current literature on intra-EU trade and the reduction of trade costs from the creation of the EU is surprisingly thin, with a lot of focus on the impacts of the Euro on trade and the effects of trade costs on countries outside of the parameters of this paper. I will focus on aggregate trade levels from 1989-1998 looking for a difference in aggregate trade levels due to the Treaty of Maastricht in 1993. The Maastricht Treaty creates lower trade costs that led to increased trade in certain trade communities. As far as I can tell the link between a decrease in trade costs and increase in aggregate trade in Europe in 1993 has not been stated explicitly.

Data:

The sources I used for my panel data include the UN Comtrade Database for aggregating trade across five European countries, the Centre d'Etudes Prospectives et d'Informations Internationales (CEPII database) for control variables of distance between countries, distance between capital cities in countries, contiguity, and two weighted measurements of distance taking into account geographic distribution of population between two countries, and finally the World Bank for data on the five countries' GDPs and populations. One observation in my dataset is all trade of a particular industry between exporter A and importer B in year Y. There are 24,631 observations over the 10-year period I observed. Industries are denoted by the standard international trade classification (SITC) codes using revision 2. I use the two-digit SITC code to specify industries. Of significance is the coal and the steel industries. These two industries will formulate my control group in the difference in differences model I run. All other industries will be in the treatment group.

Empirical Model:

This paper focuses on the Maastricht Treaty's impact on the five largest and wealthiest economies in the EU. I do not observe all the countries in the EU due to the size of the data necessary and the unwieldy nature of the UN Comtrade database with larger sample sizes of countries. Even with these five countries, the United Kingdom, France, Italy, Germany, and Spain, I can still extract valuable conclusions about the impact the treaty had on Europe and the EU for a number of reasons. These five countries are fairly representative of countries in the E.U. The United Kingdom, Germany, and France are wealthier countries in northern Europe, and have actually stood to gain less from the creation of the EU compared to other southern countries that are comparatively poorer like Spain and Italy. All five countries also speak different languages, and differ in both contiguity, size, and distance from each other. They are also some of the most dominant and interconnected economies in Europe and trade the most with other countries. Lastly, and most importantly, they all are reliable reporters of trade data from intra-EU trade. I chose to look at a ten-year period so that I have five years before and after the Maastricht Treaty took effect. This allows enough time for any non-immediate changes to be taken into account, as well as allowing me to get a sense of where the levels of aggregate trade were on average before the Maastricht Treaty took effect. The controls I use help to incorporate the gravity trade model as I take into account distance and economic size of countries. I focus on aggregate trade so I see how goods that are already part of an international trade agreement and other goods that were not part of an international trade agreement behave before and after the Maastricht Treaty takes effect. The scale of total aggregate trade is also important as I am interested in how the treaty effected total trade across the EU.

I will be running a fixed effects difference in differences model before and after the Maastricht Treaty took effect looking at the levels of aggregate trade in the coal and steel

community and all other trade goods. The coal and steel community makes an ideal control group because it was already protected from trade tariffs before the Maastricht Treaty took effect thanks to the EEC. The model I will be estimating is as follows:

$$Y_{ijst} = \beta_0 + \beta_1 * (Post * Group) + \alpha_{ij} + \alpha_t + v_{it}$$

Where α_{ij} represents the importer-exporter pair and α_t is year fixed effects. My outcome variable is in terms of importer and exporter pair (ij) year (t), and industry (s). *Post* is a dummy variable that indicates if the EU has been formed or not (1 if yes, 0 if not) and *Group* indicates if the tradable good is the control (coal or steel) or the treatment (all other goods). I want to see how the implementation of the Maastricht treaty effected intra-EU aggregate trade between partner pairs (I,j). I use robust standard errors in all regressions. My model does have a slight aggregation bias, as the data is yearly. However, the Maastricht Treaty took effect in November 3rd, leaving only two months until the start of the new year.

In terms of difference in differences assumptions, I will address the first two (Ashenfelter's dip and common trends) in the results section when I present my raw data. The third assumption, the inability for entities to change groups, is near impossible thanks to the structure of groups in my study. The below summary of statistics recognizes types of variables, their means and standard errors. The large number of observations comes from measuring all trade of a particular tradable good (quantified using the revision 3 SITC code) across ten years and five countries. The mean value of *Year* simply notes the mean value of the time interval I have chosen. The *Scaled Trade Value* is in terms of millions of dollars.

Table 1

Variable	Obs	Var type	Mean	Std. Dev.
Year	1989-1998	Interval	1993.848	2.727643
Industry	66 categories	Categorical		
Trade Value in Dollars	24,631	Continuous	3.14E+08	6.72E+08
Border	24,631	Dummy	0.308027	0.461687
EU	24,631	Dummy	0.543665	0.4981
Trade Value in 2010 Dollars	24,631	Continuous	4.61E+08	9.72E+08
Scaled Trade Value	24,631	Continuous	461	972
Logged Trade Value	24,631	Continuous		
Importer	France, Germany, Spain, Italy, UK.	Categorical		
Exporter	France, Germany, Spain, Italy, UK	Categorical		
Importer GDP	24,633	Continuous	1.83E+12	5.94E+11
Importer pop	24,633	Continuous	5.82E+07	1.29E+07
Exporter GDP	24,633	Continuous	1.84E+12	5.92E+11
Exporter pop	24,633	Continuous	5.83E+07	1.29E+07
Distance Between Partners	24,631	Continuous	1059.867	364.7784
Distance Between Partner's Capitals	24,631	Continuous	1135.968	379.1834
Weighted Distance 1	24,631	Continuous	1110.208	307.4302
Weighted Distance 2	24,631	Continuous	998.6833	329.8016

Results

Figure 1



Figure 2

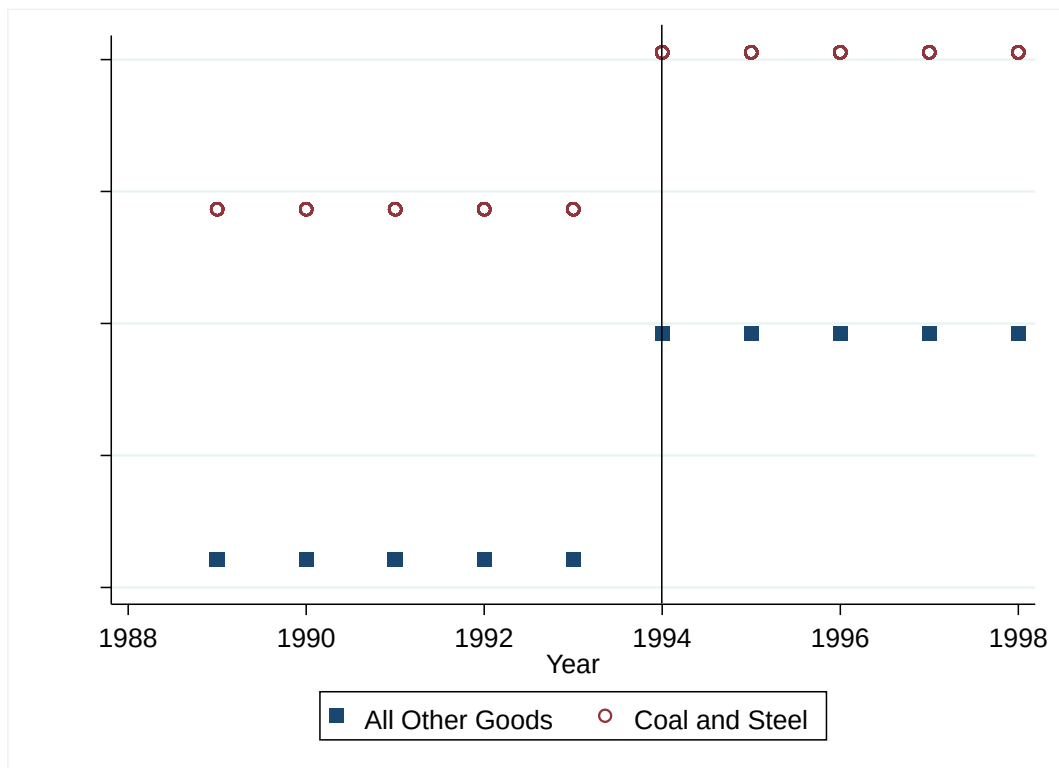


Figure 1 plots years against aggregate trade levels in millions. The vertical line indicates the first full year the Maastricht Treaty was in effect. In both periods, the control group is coal and steel. These goods had already been protected by an international trade agreement before the treaty took effect. The treatment group is all other goods traded. Figure 2 shows a simple difference in differences model with predicted values. Again, aggregate trade values are in millions and the vertical line indicates the first full year the Maastricht Treaty took effect. The control group of goods (coal and steel) is also on the top in both graphs.

Looking at figure 1 in the period before the Maastricht Treaty took effect, we observe that both the treatment and the control groups are moving together. This is significant because it reveals that we have common trends, an assumption necessary to meet in order to perform a difference in differences model. Observing the immediate years before and after the treaty took effect, we do not observe either a dip in a group before the treaty took effect nor a dramatic increase immediately after the treaty took effect. These combined effects would point to an Ashenfelter's dip. The absence of this dip, combined with common trends in both periods means that we meet another assumption necessary in order to use a difference in differences model. The last assumption, the impossibility of entities changing groups, is already met as discussed previously in the empirical model section.

Figure 2 clearly shows a larger increase in predicted levels of aggregate trade for the treated group than the control group. Before, the treated group started at 410 million dollars and increases to 495.5 million dollars after the Maastricht Treaty takes effect. That's an increase of 85.5 million dollars. Even though this difference may not be statistically significant in the simple

Table 2

	(1) Scaled Trade Value	(2) Scaled...	(3) Scaled...	(4) Scaled...	(5) Scaled...	(6) Scaled...
EU	84.83*** (6.92)	102.9*** (-6.59)	8.350 (0.43)	59.43 (1.06)	51074.0 (0.93)	
Scaled Importer GDP		51.36*** (7.32)	2.044 (0.23)			
Scaled Importer pop		-10.62** (-3.26)	9.740*** (2.33)			
Scaled Exporter GDP		66.21*** (10.06)	16.70** (1.90)			
Scaled Exporter pop		-14.42** * (-4.57)	6.029* (1.46)			
Distance Between Partners			0.440*** (6.39)			
Distance Between Partner's Capital			-0.216** * (-3.32)			
Weighted Distance 1			-1.592** * (-4.66)			
Weighted Distance 2			1.062*** (3.37)			
Contiguity			71.54*** (2.39)			
Treatment				-132.7*** (-3.21)	-132.1*** (-3.20)	
Treat*EU				26.05 (0.45)	-25454.3 (-0.46)	.2767*** (2.43)
Year					27.47*** (5.40)	
EU*Year					-25.62 (-0.93)	
EU*Year*Treatment					12.77 (0.46)	
Constant	414.5*** (49.37)	-180.8** (-1.90)	-342.9** (-1.83)	543.3*** (13.45)	-54154.5*** (-5.35)	4.43*** (77.69)
N	24631	24631	24631	24631	24631	24631

t statistics in parentheses

=** p<0.1

** p<0.05

*** p<0.01"

difference in differences model, this could be due to a small sample size. We will also see that controlling for fixed effects does yield a statistically significant model later.

Table 1 (above) summarizes my final results, starting with a simple regression with just the Treaty's existence as the explanatory variable (1) to a fixed effects difference in differences model (6). Of note is that a regression run with only GDP and population controls (2) is significant, another regression with more controls (3) is not. We don't see any significance in the simple difference in differences model or time trends model (4 and 5), but we do see significance with my final model.

The final results of the fixed effects difference in differences model are statistically significant. β_1 in my model was 0.277, indicating that after the treaty took effect there was a 27.7% increase in aggregate trade among the effected goods. I can compare this change in trade in percent terms because before taking the final results of my fixed effects difference in differences I logged the scaled value of aggregate trade (my Y variable). The Maastricht Treaty increased aggregate trade of goods not protected by trade agreements between EU member states. Worth noting is my 95% confidence interval, from .053 - .500. This indicates I am 95% sure that the true change in trade after the Maastricht Treaty took effect was between 5.3% increase to a 50% increase in trade. The reason I have such a large interval is because of large standard errors. While there are still threats to this model from endogeneity, the increase in aggregate trade is still relevant. The fact that trade among goods without previous trade barriers continued to increase is also worth noting. However, as we get farther from the treaties implementation, the risk of other outside variables impacting the increase in trade rises.

Endogeneity Concerns

As always there is a significant concern about endogeneity in my model. Addressing some of the most serious concerns, a historical analysis of Europe in the 1990s leaves relatively few and insignificant interferences with international trade during the immediate period when the treaty took effect. The creation of the European common market and the gradual implementation of the Euro could both impact aggregate trade levels. The creation of the common market has a definitive start date and I anticipate being able to properly attribute any significant change in trade during this period in my results to the Common Market rather than the Maastricht Treaty. The gradual process of implementing the Euro means that there was no common currency between member countries during the years examined in my research. Controlling for exchange rates would thus be a valid concern. However, during this period, the European countries in my study all used the European Currency Unit (ECU). The ECU was moderated by the European Exchange Rate Mechanism to help reduce exchange rate volatility and ease international trade among European Community members. I will thus assume that the exchange rate remained relatively stable between the countries in my study in terms of international trade.

One could argue that only countries with a desire to increase trade in the first place would sign the Maastricht Treaty. This would cause an issue with endogeneity as well. While this is a valid concern, I will look past it as I consider the other two pillars set up in the treaty reason enough for other countries to join as well. Also, all countries in my sample signed the treaty. Although it is impossible to eliminate all endogeneity concerns in my model, I hope some of my discussion has eased concern about the most pressing issues.

In terms of secondary results, I ran another fixed effects regression excluding any trade with Germany. Germany has the largest economy in Europe and also trades the most with other

European countries. In order to make sure that Germany alone was not causing the observed increase in trade, I removed it from the equation.

	(1)
	Trade Value (logged)
Treat*EU	0.476** (3.25)
_cons	4.203***
N	16018
t statistics in parentheses	
="* p<0.05 ** p<0.01 *** p<0.001"	

Given the above results it is clear Germany was not the lone driver of my initial results. This result then shows that the increase in trade was shared among all five countries.

Conclusion:

A curiosity of international trade and the economic function of the EU drove me to consider the impact of the Maastricht Treaty on intra-EU trade. While there are a number of papers that discuss the results of trade barriers on aggregate trade between partners, I was unable to find a specific paper that described the Maastricht Treaty's direct impact on intra-EU trade. I show that aggregate trade in goods not previously protected by international trade agreements increased dramatically once the treaty took effect. I demonstrate this using a fixed effects difference in differences model controlling for both entity and time fixed effects. The increase in trade can be attributed to the decrease in trade barriers resulting from the Maastricht Treaty. This

result is what the current literature would predict in the event of a decrease in trade barriers. I contribute a model that proves the link between a decrease in trade barriers and an increase in aggregate intra-EU trade. Given this link, it would be appropriate to assume any further decrease in trade barriers between EU members states would increase aggregate trade. Having already established internal validity, questions naturally emerge about what external validity this model may have. Given the population and setting in Europe today is similar, applying the results of this model to modern Europe is a natural next step. This is particularly important as currently, many member states in the EU are now trying to throw up trade barriers intentionally to make domestic producers more competitive against foreign ones. This model could shed light on what kind of economic impacts this could have for Europe, and how it might threaten the very ideals the European Union was founded under. The EU must weigh the impact of decreasing trade barriers on domestic firms in member states and the benefits of increased aggregate trade.

Refrances

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